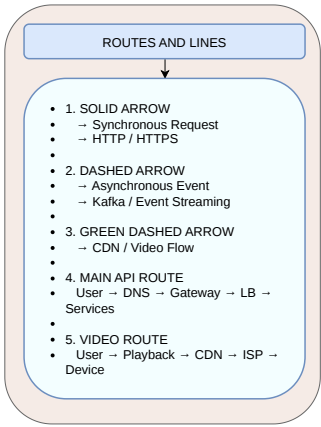
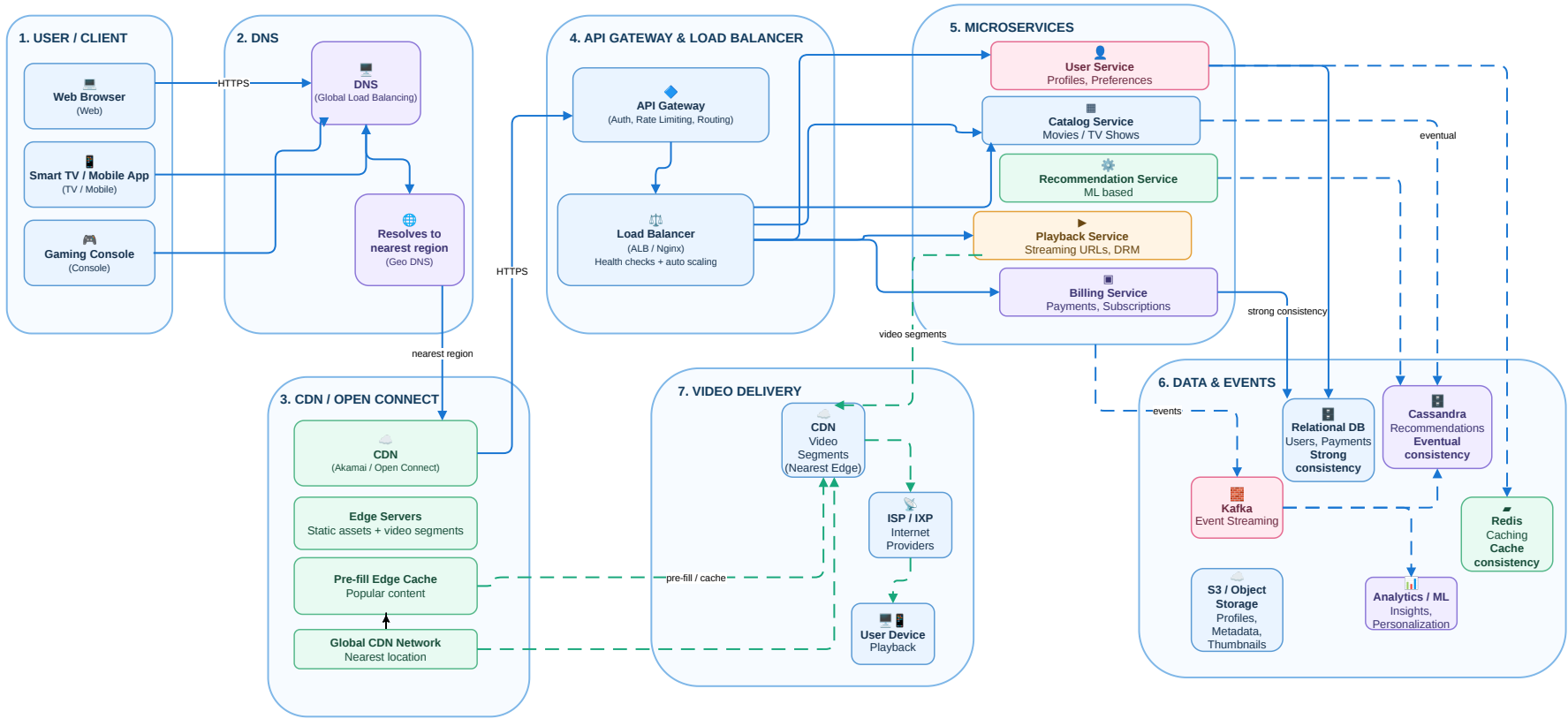
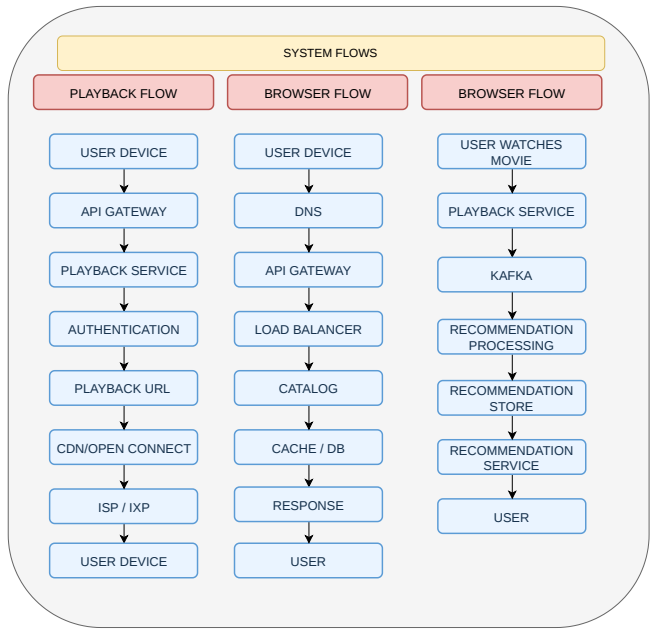


N NETFLIX SYSTEM DESIGN



▶ **PLAYBACK PATH (Video Streaming)**



FUNCTIONAL REQUIREMENTS:

- 1. USER MANAGEMENT**
 - Register / Login / Logout
 - Profiles & Parental Controls
- 2. CONTENT DISCOVERY**
 - Browse / Search / Filter
 - Genres & Languages
- 3. RECOMMENDATIONS**
 - Personalized Content
 - ML-Based Suggestions
- 4. VIDEO PLAYBACK**
 - Play / Pause / Seek / Resume
 - Subtitles & Multi-Audio
 - Adaptive Bitrate
- 5. SUBSCRIPTION**
 - Plans & Payments
 - Billing & Renewal
- 6. USER ACTIVITY**
 - Watch History
 - My List / Continue Watching
- 7. CONTENT MANAGEMENT**
 - Upload → Transcode → DRM → Publish
- 8. ANALYTICS**
 - User Events
 - Experimentation

NON FUNCTIONAL REQUIREMENTS:

- 1. AVAILABILITY**
 - 99.99% Target
 - Multi-Region & Failover
- 2. SCALABILITY**
 - Horizontal Scaling
 - Auto Scaling
- 3. PERFORMANCE**
 - Low API Latency
 - Low Playback Startup Time
- 4. RELIABILITY**
 - Fault Tolerance
 - Retry + Timeout
 - Circuit Breaker
- 5. DATA**
 - Durability
 - Replication
- 6. SECURITY**
 - Authentication
 - Authorization
 - TLS + DRM
- 7. OBSERVABILITY**
 - Logs
 - Metrics
 - Traces
- 8. RECOVERY**
 - Disaster Recovery
 - Graceful Degradation

ESTIMATION

- 1. SCALE**
 - 300M+ Memberships
 - 10% DAU = 30M Users/Day
- 2. API TRAFFIC**
 - ~20 Requests / User
 - Daily Requests = DAU × Requests
- 3. RPS**
 - Avg RPS = Daily Requests / 86,400
 - Peak RPS = Avg RPS × Peak Factor
- 4. VIDEO TRAFFIC**
 - Video Traffic >> API Traffic
 - CDN Handles Majority of Delivery
- 5. ASSUMPTION**
 - Numbers Are Illustrative
 - Not Production Netflix Data
- 6. USER DATA**
 - 300M × 20KB = 6TB